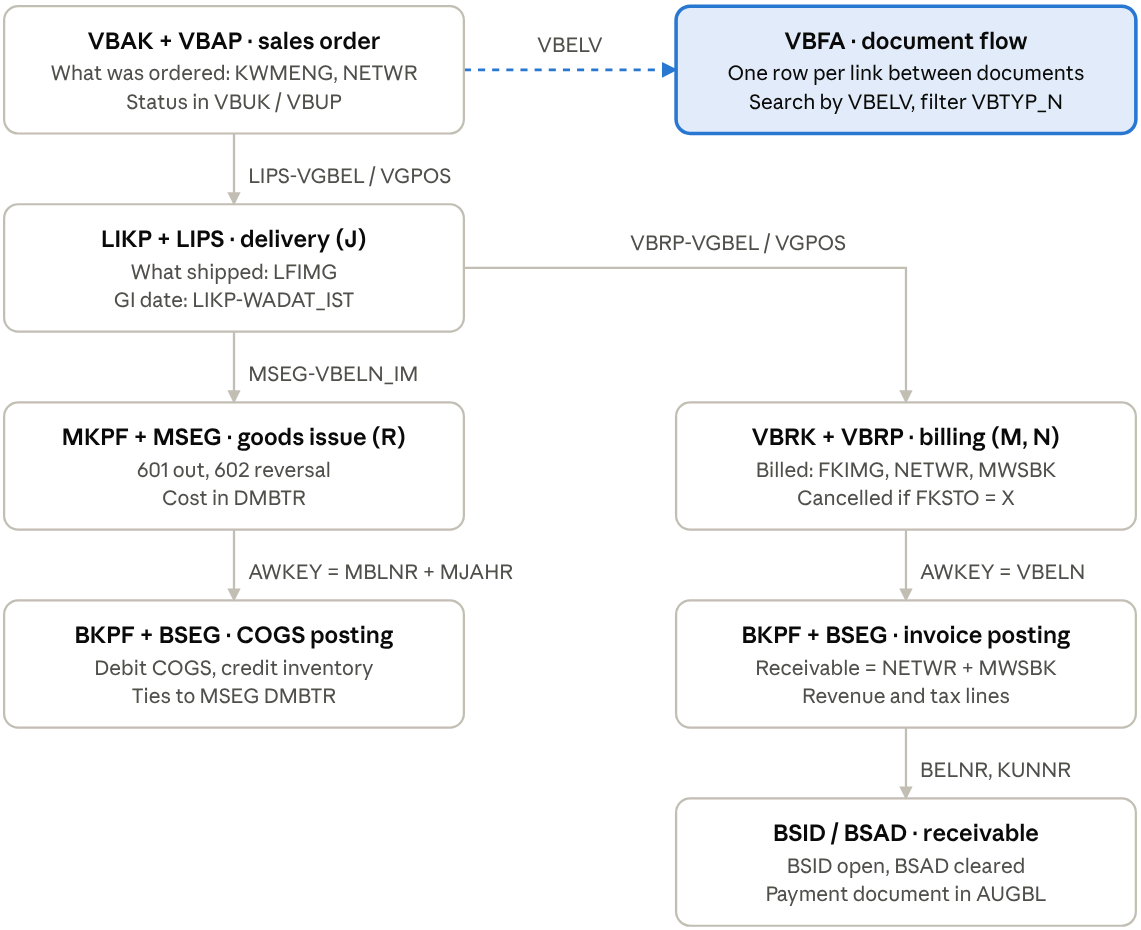
VBFA Document Flow Trace and Reconciliation (SAP ECC)

Nov 9, 2026 · @Harsh

Overview

VBFA records which document followed which, with the quantity carried over at each step; the authoritative quantities, values and statuses sit in each document's own tables. To rebuild and reconcile the flow you read VBFA for the links, then each document's own tables for the numbers.

Each SD step has its own tables; VBFA records the links



SD document chain · tables and join fields

The left column is the physical flow (order, delivery, goods issue, cost posting). The right column is the financial flow (billing, invoice posting, receivable). Each arrow is labelled with the reference field that links the two tables, so you can trace without VBFA when it is slow.

Order of work: VBAK and VBAP for the baseline, VBFA for the links, then LIPS, MSEG, VBRP and VBRK for quantities and values, then BKPF, BSEG and BSID/BSAD for FI. SQVI Query 1 and Query 2 below collapse most of this into two runs, and the reconciliation section ties it all together.

Worked example

Every step below traces this one sales order line end to end. Numbers are illustrative; your number ranges and document types will differ.

Item	Example value
Sales order	0010045678, item 000010
Customer (sold-to)	0000201234
Material	100234, store shelving unit
Ordered quantity	50 EA at $500.00 = $25,000.00 net
Sales org / company code	1000 / 1000
Tax	HST 13%

What happened to this line:

1. Delivery 0080012345 shipped 30 EA. Goods issue 4900056781 posted
2. Delivery 0080012399 shipped 20 EA. Goods issue 4900056802 posted
3. Invoice 0090034567 billed delivery 1: 30 EA, $15,000.00
4. Invoice 0090034590 billed delivery 2: 20 EA, $10,000.00, but with the wrong billing date
5. Invoice 0090034590 was cancelled by 0090034595
6. Invoice 0090034601 re-billed delivery 2: 20 EA, $10,000.00

The goal: rebuild this flow from tables, then prove ordered = delivered = goods issued = billed (net of the cancellation) = posted to FI.

Step 1: VBAK, VBAP, VBUK, VBUP, the order

Start with what was ordered and what SAP thinks is still open. These four tables are your baseline for every reconciliation later.

VBAK: order header

In SE16N, paste your order list into `VBELN`.

VBELN	AUART	VKORG	KUNNR	ERDAT	NETWR	WAERK	KNUMV
0010045678	OR	1000	0000201234	2026-06-02	25,000.00	CAD	0000876543

- `AUART` = order type (OR standard order, RE returns, CR credit memo request)
- `KNUMV` = pricing condition number. Use it in `KONV` if you need price details

VBAP: order items

Same `VBELN` list.

VBELN	POSNR	MATNR	ARKTX	KWMENG	VRKME	NETPR	NETWR	WERKS	ABGRU
0010045678	000010	100234	Store shelving unit	50	EA	500.00	25,000.00	1025	

- `KWMENG` = ordered quantity, your starting number for the quantity check
- `ABGRU` = rejection reason. A filled value means the line was rejected and should be excluded from reconciliation

VBUK / VBUP: status

VBUK is header status, VBUP is item status. Paste the same order list.

Field (VBUP)	Example	Meaning
`LFSTA`	C	Delivery status
`WBSTA`	C	Goods issue status (on the delivery items)
`FKSTA`	C	Delivery-related billing status
`FKSAA`	blank	Order-related billing status (only for order-billed items)

Field (VBUP)	Example	Meaning
`GBSTA`	C	Overall processing status

Values: **A** = not processed, **B** = partially processed, **C** = completely processed, blank = not relevant.

Status is SAP's own verdict. Your reconciliation should agree with it: any line at C must tie out, any line at A or B is an open item to explain.

Step 2: VBFA, the forward trace

VBFA stores one row per link between a preceding document and a subsequent one. Search it by the **preceding** document (`VBELV`), never by the subsequent one: `VBELV` is the start of the table key and is fast, `VBELN` is not.

Key fields

Field	Meaning
`VBELV` / `POSNV`	Preceding document and item (what you paste)
`VBELN` / `POSNN`	Subsequent document and item (what you find)
`VBTYP_V` / `VBTYP_N`	Category of preceding / subsequent document
`RFMNG` / `MEINS`	Quantity carried over, and its unit
`RFWRT` / `WAERS`	Value carried over, and its currency
`PLMIN`	+ or - : direction of the quantity. Cancellations and returns carry -
`ERDAT`	Date the link was created
`MJAHR`	Year of a material document (goods issue rows only)

In SE16N

1. Table `VBFA`
2. `VBELV` : paste your order numbers
3. `VBTYP_N` : leave blank on the first run so you see everything, then filter per step

4. Clear "Maximum No. of Hits", execute

Example output for order 0010045678

VBELV	POSNV	VBELN	POSNN	VBTYP_N	RFMNG	RFWRT	PLMIN
0010045678	000010	0080012345	000010	J	30	0.00	+
0010045678	000010	0080012399	000010	J	20	0.00	+
0010045678	000010	0090034567	000010	M	30	15,000.00	+
0010045678	000010	0090034590	000010	M	20	10,000.00	+
0010045678	000010	0090034595	000010	N	20	10,000.00	-
0010045678	000010	0090034601	000010	M	20	10,000.00	+

Reading it

- `J` = delivery, `M` = invoice, `N` = invoice cancellation, `R` = goods movement. Full list in the cheat sheet
- VBFA under an order can list later documents too (invoices, sometimes goods issues), not only the next step. Always filter `VBTYP_N` before summing, or you add delivery and invoice quantities together
- Signed quantity = `RFMNG` × (+1 if `PLMIN` is + or blank, -1 if -). Billed net = 30 + 20 - 20 + 20 = 50 EA
- Delivery rows carry no value (`RFWRT` 0). Take values from billing, not from the delivery link

Second pass: from the deliveries

Paste the delivery numbers (the `J` rows) into `VBELV` and filter `VBTYP_N` = R to get the goods issues.

VBELV	VBELN	VBTYP_N	RFMNG	PLMIN	MJAHR
0080012345	4900056781	R	30	+	2026
0080012399	4900056802	R	20	+	2026

Export both passes to a sheet called `VBFA`.

Step 3: LIKP, LIPS, MKPF, MSEG, deliveries and goods issue

LIPS tells you what was actually picked and shipped; MSEG tells you what left inventory and at what cost. Use the delivery and material document numbers from VBFA.

LIKP: delivery header

Paste the `J` document numbers into `VBELN`.

VBELN	LFART	VSTEL	KUNNR	LFDAT	WADAT_IST
0080012345	LF	1025	0000201234	2026-06-10	2026-06-10
0080012399	LF	1025	0000201234	2026-06-24	2026-06-24

- `WADAT_IST` = actual goods issue date. Blank means goods have not left yet
- `KUNNR` here is the ship-to, which can differ from the sold-to on the order

LIPS: delivery items

Same delivery numbers.

VBELN	POSNR	MATNR	LFIMG	VRKME	VGBEL	VGPOS	WERKS	LGORT
0080012345	000010	100234	30	EA	0010045678	000010	1025	0001
0080012399	000010	100234	20	EA	0010045678	000010	1025	0001

- `LFIMG` = delivered quantity in sales unit
- `VGBEL` / `VGPOS` = the order and item this delivery came from. This is the reference field you use for backward traces and SQVI joins
- Batch-managed materials may show extra sub-items (`UECHA` filled). Sum only the main items or only the sub-items, not both

MKPF: material document header

Paste the `R` document numbers into `MBLNR`, with `MJAHR` = 2026.

MBLNR	MJAHR	BUDAT	XBLNR	USNAM
4900056781	2026	2026-06-10	0080012345	WHUSER01
4900056802	2026	2026-06-24	0080012399	WHUSER01

`XBLNR` normally holds the delivery number, a quick cross-check.

MSEG: material document items

Same `MBLNR` and `MJAHR`.

MBLNR	ZEILE	BWART	MATNR	MENGE	MEINS	DMBTR	WERKS	VBELN_IM
4900056781	0001	601	100234	30	EA	10,800.00	1025	0080012345
4900056802	0001	601	100234	20	EA	7,200.00	1025	0080012399

- `BWART` 601 = goods issue for delivery, 602 = its reversal. Net them: 601 minus 602
- `MENGE` is in base unit (`MEINS`), which can differ from the sales unit in LIPS. Convert before comparing
- `DMBTR` = inventory value at cost. This becomes cost of goods sold in FI (Step 5)

Step 4: VBRK and VBRP, billing and cancellations

VBRK and VBRP hold what was billed and whether it reached accounting. The main trap here is cancellations: amounts in VBRK are stored as positive numbers, so the sign comes from the document category.

VBRK: billing header

Paste the `M`, `N`, `O` and `P` document numbers from VBFA into `VBELN`.

VBELN	FKART	VBTYP	FKDAT	BUKRS	KUNRG	NETWR	MWSBK	RFBSK	FKSTO	SFAKN
0090034567	F2	M	2026-06-10	1000	0000201234	15,000.00	1,950.00	C		
0090034590	F2	M	2026-07-24	1000	0000201234	10,000.00	1,300.00	C	X	
0090034595	S1	N	2026-07-24	1000	0000201234	10,000.00	1,300.00	C		0090034590
0090034601	F2	M	2026-06-24	1000	0000201234	10,000.00	1,300.00	C		

- `FKSTO` = X means this invoice has been cancelled
- `SFAKN` sits on the cancellation document and points to the invoice it cancelled
- `RFBSK` = posting status. C = posted to accounting, A = blocked for posting, blank or B = not (fully) posted. Anything not C will have no FI document yet
- `KUNRG` = payer, the customer the receivable is posted to
- `NETWR` = net value, `MWSBK` = tax. Gross = NETWR + MWSBK

Two ways to get the valid billed amount (both should agree)

1. **Exclude:** keep only `VBTYP` M with `FKSTO` blank, plus credit and debit memos. Here: 15,000 + 10,000 = 25,000
2. **Net with signs:** M and P count +, N and O count -. Here: 15,000 + 10,000 - 10,000 + 10,000 = 25,000

Do not mix the two methods. Excluding the cancelled invoice and also subtracting its cancellation double-counts the reversal.

VBRP: billing items

Same document list.

VBELN	POSNR	FKIMG	VRKME	NETWR	MWSBP	WAVWR	VGBEL	AUBEL	AUPOS
0090034567	000010	30	EA	15,000.00	1,950.00	10,800.00	0080012345	0010045678	000010
0090034590	000010	20	EA	10,000.00	1,300.00	7,200.00	0080012399	0010045678	000010
0090034595	000010	20	EA	10,000.00	1,300.00	7,200.00	0080012399	0010045678	000010
0090034601	000010	20	EA	10,000.00	1,300.00	7,200.00	0080012399	0010045678	000010

- `FKIMG` = billed quantity
- `VGBEL` = the delivery billed, `AUBEL` / `AUPOS` = the original order and item. Both are reference fields you can join on directly
- `WAVWR` = cost carried on the billing item. It should match the MSEG `DMBTR` for the same delivery

Step 5: BKPF, BSEG, BSID, BSAD, accounting and receivables

Every posted invoice and every goods issue creates an FI document. Find them through `BKPF-AWKEY`, not by assuming the FI number equals the SD number (it often does, but not always).

BKPF: find the FI documents

Source	AWTYP	AWKEY =	Example AWKEY
Billing document	VBRK	Billing number	0090034567
Goods issue	MKPF	Material doc number + year	49000567812026

In SE16N, always fill `BUKRS` = 1000 and `GJAHR` = 2026, then paste the AWKEY list into `AWKEY`.

BELNR	BLART	AWTYP	AWKEY	BUDAT	XBLNR
0090034567	RV	VBRK	0090034567	2026-06-10	0090034567
0090034590	RV	VBRK	0090034590	2026-07-24	0090034590
0090034595	RV	VBRK	0090034595	2026-07-24	0090034595
0090034601	RV	VBRK	0090034601	2026-06-24	0090034601
4900056781	WL	MKPF	49000567812026	2026-06-10	0080012345
4900056802	WL	MKPF	49000568022026	2026-06-24	0080012399

BSEG: the FI lines

Same `BUKRS`, `GJAHR`, paste the FI `BELNR` list. Example for invoice 0090034567 and its goods issue:

BELNR	BUZEI	KOART	HKONT / KUNNR	SHKZG	DMBTR	MWSKZ
0090034567	1	D	0000201234	S	16,950.00	
0090034567	2	S	400000 Revenue	H	15,000.00	O1
0090034567	3	S	220000 HST payable	H	1,950.00	O1
4900056781	1	S	500000 COGS	S	10,800.00	
4900056781	2	S	140000 Inventory	H	10,800.00	

- `KOART` D = customer line, the receivable. Its `DMBTR` must equal VBRK `NETWR` + `MWSBK`
- Revenue lines must equal VBRK `NETWR`. Tax lines must equal `MWSBK`
- COGS on the goods issue must equal MSEG `DMBTR`
- The cancellation 0090034595 posts the exact mirror of 0090034590 (D on H, revenue and tax on S)

BSID / BSAD: is it paid?

BSID holds open customer items, BSAD holds cleared ones. Filter `BUKRS` = 1000, `KUNNR` = 0000201234, and paste the invoice `BELNR` list.

Table	BELNR	DMBTR	SHKZG	AUGDT	AUGBL
BSAD	0090034567	16,950.00	S	2026-07-10	1400005501
BSAD	0090034590	11,300.00	S	2026-07-24	0090034595
BSAD	0090034595	11,300.00	H	2026-07-24	0090034595

Table	BELNR	DMBTR	SHKZG	AUGDT	AUGBL
BSID	0090034601	11,300.00	S		

- `AUGBL` = clearing document. A payment (here 1400005501) or the cancellation clearing its own invoice
- The cancelled pair should clear each other. If it sits in BSID, someone has to clear it manually
- An invoice missing from both tables was never posted. Check VBRK `RFBSK`

Backward trace: starting from an invoice, delivery or FI document

When your Excel list holds invoices or FI documents instead of orders, walk back through reference fields. Searching VBFA by the subsequent document (`VBELN`) works, but it is not on the table key and can time out on a large system.

You have	Table	Filter on	Read	Gives you
FI document number	BKPF	`BUKRS` , `GJAHR` , `BELNR`	`AWTYP` , `AWKEY`	Billing number (VBRK) or material doc + year (MKPF)
Customer line in FI	BSEG	`BUKRS` , `GJAHR` , `BELNR`	`VBELN` on the D line	Billing number
Invoice	VBRP	`VBELN`	`VGBEL` / `VGPOS`	Delivery and item
Invoice	VBRP	`VBELN`	`AUBEL` / `AUPOS`	Order and item, directly
Cancellation	VBRK	`VBELN`	`SFAKN`	The invoice it cancelled
Delivery	LIPS	`VBELN`	`VGBEL` / `VGPOS`	Order and item
Goods issue	MSEG	`MBLNR` , `MJAHR`	`VBELN_IM` / `VBELP_IM`	Delivery and item
Order	VBAP	`VBELN`	`VGBEL` / `VGPOS`	Quotation or contract it was copied from

Example: from FI document 0090034601 back to the order

1. BKPF: `BELNR` 0090034601 shows `AWTYP` VBRK, `AWKEY` 0090034601, so the billing document is 0090034601
2. VBRP: `VBELN` 0090034601 shows `VGBEL` 0080012399 (delivery) and `AUBEL` 0010045678 (order)
3. VBRK: `FKSTO` is blank, so this invoice is still valid
4. Now run the forward trace from order 0010045678 to see everything else on that order, including the earlier cancelled invoice

Rule of thumb: go backward with reference fields to find the order, then go forward with VBFA from the order to get the complete picture.

SQVI queries

These two queries replace most of Steps 1 to 4 with one run each. They join on reference fields (`VGBEL` , `VGPOS`), not on VBFA, so there is no double counting from VBFA's multi-level rows.

Query 1: order to delivery to billing

Create

1. `SQVI` , name `ZSD_FLOW` , **Create**
2. Title `Order Delivery Billing Flow` , **Table join**, **Basis mode**
3. **Edit > Insert table**: `VBAK` , `VBAP` , `LIPS` , `VBRP` , `VBRK`

Joins (delete every auto link first)

From	To	Join type
`VBAK-VBELN`	`VBAP-VBELN`	Inner
`VBAP-VBELN`	`LIPS-VGBEL`	Left outer
`VBAP-POSNR`	`LIPS-VGPOS`	Left outer
`LIPS-VBELN`	`VBRP-VGBEL`	Left outer
`LIPS-POSNR`	`VBRP-VGPOS`	Left outer
`VBRP-VBELN`	`VBRK-VBELN`	Left outer

Set left outer by right-clicking the link line. Once a table is left outer, every table after it in the chain must be left outer too, or SAP rejects the join. Left outer is what keeps undelivered and unbilled order lines in the output, which is exactly what you want to find.

Selection fields

`VBAK-VBELN`, `VBAK-AUART`, `VBAK-VKORG`, `VBAK-KUNNR`, `VBAK-ERDAT`

List fields

Table	Fields
VBAK	`VBELN`, `AUART`, `KUNNR`
VBAP	`POSNR`, `MATNR`, `KWMENG`, `NETWR`, `ABGRU`
LIPS	`VBELN`, `POSNR`, `LFIMG`
VBRP	`VBELN`, `POSNR`, `FKIMG`, `NETWR`, `MWSBP`
VBRK	`VBTYP`, `FKART`, `FKDAT`, `FKSTO`, `SFAKN`, `RFBSK`

Several tables have a field called `VBELN`. Rename the column headers in Excel right after export (Order, Delivery, Invoice) so you don't mix them up.

Run: F8, paste order numbers into `VBELN`, execute, export as sheet `FLOW`.

Example output

Order	Item	KWMENG	Delivery	LFIMG	Invoice	FKIMG	VBRP NETWR	VBTYP	FKSTO
0010045678	000010	50	0080012345	30	0090034567	30	15,000.00	M	
0010045678	000010	50	0080012399	20	0090034590	20	10,000.00	M	X
0010045678	000010	50	0080012399	20	0090034595	20	10,000.00	N	
0010045678	000010	50	0080012399	20	0090034601	20	10,000.00	M	

Order and delivery quantities repeat on every invoice row. Never sum `KWMENG` or `LFIMG` from this sheet directly; the reconciliation section shows how.

Limit: order-related billing (services, billed straight from the order with no delivery) will show no invoice here. For those, join `VBAP` to `VBRP` on `VBRP-AUBEL` and `VBRP-AUPOS` in a separate query.

Query 2: delivery to goods issue with cost

1. `SQVI`, name `ZSD_GI`, Table join, Basis mode
2. Tables: `LIKP`, `LIPS`, `MSEG`
3. Joins: `LIKP-VBELN` to `LIPS-VBELN` (inner), `LIPS-VBELN` to `MSEG-VBELN_IM` and `LIPS-POSNR` to `MSEG-VBELP_IM` (both left outer)
4. Selection: `LIPS-VGBEL` (paste the order list), `MSEG-BWART`
5. List: `LIKP-VBELN`, `WADAT_IST`, `LIPS-POSNR`, `VGBEL`, `VGPOS`, `LFIMG`, `MSEG-MBLNR`, `MJAHR`, `BWART`, `MENGE`, `MEINS`, `DMBTR`
6. Run with `BWART` 601 and 602, export as sheet `GI`

A delivery row with no MSEG match means goods were never issued.

Reconciliation

Reconcile at two levels: per order item (quantity and value through the chain) and per billing document (SD to FI). The example ties out at both.

The six checks

#	Check	Compare	Example
1	Ordered vs delivered	VBAP `KWMENG` vs sum LIPS `LFIMG`	50 = 30 + 20
2	Delivered vs goods issued	LIPS `LFIMG` vs MSEG `MENGE` (601 minus 602)	50 = 30 + 20
3	Goods issued vs billed	MSEG qty vs VBRP `FKIMG` net of cancellations	50 = 30 + 20 - 20 + 20
4	Order value vs billed value	VBAP `NETWR` vs VBRP `NETWR` net	25,000 = 25,000
5	Billing vs FI receivable	VBRK `NETWR` + `MWSBK` vs BSEG D line	16,950 = 16,950 per invoice
6	Cost: GI vs billing	MSEG `DMBTR` vs VBRP `WAVWR` net	18,000 = 18,000

Workbook layout

Rearrange each export so its columns match, or adjust the letters in the formulas.

Sheet	Source	Columns used
`ITEMS`	VBAP	A Order, B Item, D KWMENG, H NETWR
`LIPS`	LIPS	A Delivery, C LFIMG, D VGBEL, E VGPOS
`GI`	SQVI Query 2	D VGBEL, E VGPOS, I BWART, J MENGE, L DMBTR
`FLOW`	SQVI Query 1	A Order, B Item, G FKIMG, H Net, J VBTYP
`BILL`	VBRK	A VBELN, B VBTYP, C NETWR, D MWSBK, E RFBSK
`FI`	BKPF + BSEG D lines	A AWKEY, C DMBTR

Helper columns on `FLOW` (row 2, fill down)

Col	Header	Formula
L	Key	`=A2&"-"&B2`
M	Sign	`=IF(OR(J2="N",J2="O"),-1,1)`
N	Signed qty	`=G2*M2`
O	Signed net	`=H2*M2`

Item-level checks on `ITEMS` (row 2, fill down)

Col	Header	Formula
C	Key	`=A2&"-"&B2`
E	Delivered	`=SUMIFS(LIPS!C:C,LIPS!D:D,A2,LIPS!E:E,B2)`
F	Goods issued	`=SUMIFS(GI!J:J,GI!D:D,A2,GI!E:E,B2,GI!I:I,"601")-SUMIFS(GI!J:J,GI!D:D,A2,GI!E:E,B2,GI!I:I,"602")`
G	Billed qty	`=SUMIFS(FLOW!N:N,FLOW!L:L,C2)`
I	Billed value	`=SUMIFS(FLOW!O:O,FLOW!L:L,C2)`

Col	Header	Formula
J	Status	`=IF(D2<>E2,"Not fully delivered",IF(E2<>F2,"GI missing",IF(F2<>G2,"Not fully billed",IF(ABS(H2-I2)>0.01,"Value diff","Ties out"))))`

Delivered and goods issued come from LIPS and GI, not from `FLOW`, because `FLOW` repeats delivery quantities on every invoice row.

Example result

Order	Item	Ordered	Delivered	Goods issued	Billed qty	Order value	Billed value	Status
0010045678	000010	50	50	50	50	25,000.00	25,000.00	Ties out

Billing-level check on `BILL` (row 2, fill down)

Col	Header	Formula
F	Gross	`=C2+D2`
G	FI receivable	`=SUMIFS(FI!C:C,FI!A:A,A2)`
H	Check	`=IF(E2<>"C","Not posted",IF(ABS(F2-G2)<0.01,"OK","Diff "&TEXT(F2-G2,"#,##0.00")))`

VBELN	VBTYP	Gross	FI receivable	Check
0090034567	M	16,950.00	16,950.00	OK
0090034590	M	11,300.00	11,300.00	OK
0090034595	N	11,300.00	11,300.00	OK
0090034601	M	11,300.00	11,300.00	OK

Extra audit checks

- **Cutoff:** compare billing date (VBRK `FKDAT`) with goods issue date (LIKP `WADAT_IST`). Different months means revenue may sit in the wrong period. In the example, invoice 0090034590 was dated July for a June shipment, which is why it was cancelled and re-billed
- **Status vs your numbers:** every item with VBUP `GBSTA` = C should show "Ties out". A C status with a difference, or an A/B status that ties out, needs a look
- **Open receivables:** valid invoices still in BSID past their due date
- **Unposted billing:** VBRK `RFBSK` other than C means revenue is not in the ledger yet
- **Rejected lines:** items with VBAP `ABGRU` filled should have nothing delivered or billed

Troubleshooting

Symptom	Likely cause	Fix
VBFA returns nothing for an order	Leading zeros missing (10045678 vs 0010045678)	Paste numbers as text with all 10 digits, or let SE16N convert by typing one manually to test
Billed quantity is double the order	Summed VBFA rows without filtering `VBTYP_N`, so delivery and invoice rows were added together	Filter `VBTYP_N` per check
Billed value is short by one invoice	Excluded the cancelled invoice and also subtracted its cancellation	Use one method only: exclude, or net with signs
Billed value too high	Cancellation (N) or credit memo (O) counted as positive	Apply the sign column from the reconciliation section
VBFA search by `VBELN` times out	Subsequent document is not on the table key	Go backward with VBRP `VGBEL` / `AUBEL` or LIPS `VGBEL` instead
SUMIFS returns 0	Item number is text (000010) in one sheet and a number (10) in another	Convert both columns to text, or both to numbers

Symptom	Likely cause	Fix
GI qty doesn't match delivered	MSEG in base unit, LIPS in sales unit (for example CS vs EA)	Compare LIPS `LGMNG` (delivery qty in base unit) to MSEG `MENGE`
Delivered qty looks doubled	Batch split: main item plus sub-items with `UECHA` filled	Sum main items only (`UECHA` blank) or sub-items only
Invoice has no FI document	Not released to accounting	Check VBRK `RFBSK`. A = posting block, blank or B = not posted
FI amount differs from billing	Rounding, or a manual FI adjustment against the invoice	Look at BSEG lines beyond the first three, and at later documents clearing it
Order line shows nothing delivered or billed	Rejected line or still open	Check VBAP `ABGRU` and VBUP `LFSTA`
Service order shows no invoice in Query 1	Order-related billing, no delivery in between	Join VBAP to VBRP on `AUBEL` / `AUPOS`
Returns mixed in with sales	Returns order (RE) flows the opposite way: returns delivery `T`, credit memo `O`	Filter `AUART` or reconcile returns separately with the sign flipped

Authorization: SD and FI tables are often split across roles. If one table says "no authorization", note it and ask for display access rather than working around it.

Cheat sheet

Tables and join keys

Table	Holds	Filter on	Joins to
VBAK	Order header	`VBELN`, `AUART`, `VKORG`, `ERDAT`	VBAP via `VBELN`
VBAP	Order items	`VBELN`, `POSNR`	LIPS via `VGBEL` / `VGPOS`, VBRP via `AUBEL` / `AUPOS`

Table	Holds	Filter on	Joins to
VBUK / VBUP	Header / item status	`VBELN` (+ `POSNR`)	Any SD document
VBPA	Partners (sold-to, ship-to, payer)	`VBELN`, `PARVW`	KNA1 via `KUNNR`
KONV	Pricing conditions	`KNUMV`	VBAK / VBRK `KNUMV`
VBFA	Document flow	`VBELV`, `POSNV`, `VBTYP_N`	Any SD document
LIKP	Delivery header	`VBELN`	LIPS via `VBELN`
LIPS	Delivery items	`VBELN`, `VGBEL`	VBAP via `VGBEL` / `VGPOS`
MKPF	Material document header	`MBLNR`, `MJAHR`	BKPF via `AWKEY` = MBLNR + MJAHR
MSEG	Material document items	`MBLNR`, `MJAHR`, `BWART`	LIPS via `VBELN_IM` / `VBELP_IM`
VBRK	Billing header	`VBELN`, `FKDAT`, `BUKRS`	BKPF via `AWKEY` = VBELN
VBRP	Billing items	`VBELN`	LIPS via `VGBEL` / `VGPOS`, VBAP via `AUBEL` / `AUPOS`
BKPF	FI header	`BUKRS`, `GJAHR`, `BELNR` or `AWKEY`	BSEG via `BUKRS` + `BELNR` + `GJAHR`
BSEG	FI lines	`BUKRS`, `GJAHR`, `BELNR`	`KUNNR` customer, `VBELN` billing doc
BSID / BSAD	Customer open / cleared items	`BUKRS`, `KUNNR`, `BELNR`	Payment via `AUGBL`
KNA1	Customer master	`KUNNR`	Name, address

VBTYP document categories

Code	Document	Sign in reconciliation
A	Inquiry	n/a

Code	Document	Sign in reconciliation
B	Quotation	n/a
C	Order	n/a
E	Scheduling agreement	n/a
G	Contract	n/a
H	Returns	n/a
I	Order without charge	n/a
J	Delivery	+
K	Credit memo request	n/a
L	Debit memo request	n/a
M	Invoice	+
N	Invoice cancellation	-
O	Credit memo	-
P	Debit memo	+
Q	WMS transfer order	n/a
R	Goods movement	+ (check `PLMIN`)
S	Credit memo cancellation	+
T	Returns delivery	-
U	Pro forma invoice	Exclude, never posts to FI
5 / 6	Intercompany invoice / credit memo	Reconcile separately
8	Shipment	n/a

Other codes

Field	Value	Meaning
`BWART`	601 / 602	Goods issue for delivery / reversal
`BWART`	651	Returns receipt from customer

Field	Value	Meaning
`RFBSK`	C	Posted to accounting
`RFBSK`	A	Posting blocked
`FKSTO`	X	Billing document has been cancelled
Status (VBUP)	A / B / C	Not / partially / completely processed
`SHKZG`	S / H	Debit / credit
`KOART`	D / S / A	Customer / GL / asset line